XRPL Validator — Daily Engineering Report · 2026-07-24

Period: 2026-07-24 00:00 to 24:00 US Eastern (2026-07-24 04:00 to 2026-07-25 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-07-24): 0.99202 (172 missed of 21,553). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): CheckCash requires exactly one of Amount / DeliverMin** (`609c827`, xrpl-validator) Preflight required Amount, so it rejected every DeliverMin cash-out as temMALFORMED — a valid transaction the network accepts, which is a consensus-relevant divergence (we'd drop a tx mainnet applies). rippled's rule is an XOR: bool(Amount) == bool(DeliverMin) => temMALFORMED (CheckCash.cpp:57). Amount is a fixed cash-out; DeliverMin a floor for a partial one.
- **fix(xrpl-ledger): a Payment to a pseudo-account is tecNO_PERMISSION** (`3a718aa`, xrpl-validator) An AMM, Vault, or LoanBroker account receives value only through its own transaction type — never an ordinary Payment. rippled rejects it with tecNO_PERMISSION (Payment.cpp:635, isPseudoAccount(sleDst)), where a pseudo-account is any AccountRoot carrying a designator field: sfAMMID, sfVaultID, or sfLoanBrokerID. The engine had no such check and delivered.
- **fix(xrpl-ledger): OfferCreate crossing spends only accountFunds, not TakerGets** (`1de6d87`, xrpl-validator) rippled clamps the crossing input to the taker's actual funds -- "Don't send more than our balance", OfferCreate.cpp:399-401, where inStartBalance = accountFunds(psb, accountID_, takerAmount.in, ...) is the XRP balance minus reserve. TakerGets bounds what may REST, never what may be spent. We clamped only at TakerGets, so an underfunded offer crossed with money the account did not have.
- **fix(xrpl-ledger): honour keepRoot when a directory page empties** (`34fca11`, xrpl-validator) dir_remove always deleted an emptied root page -- rippled's keepRoot=false behaviour -- with no way to ask for anything else. Most owner-directory removals pass true, so we emitted the page as Deleted (:2) where mainnet emits it Modified (:1).

- **fix(xrpl-ledger): CheckCancel unlinks the Check from both owner directories** (`b79f40f`, xrpl-validator) A Check is linked into TWO owner directories -- the writer's via sfOwnerNode and the destination's via sfDestinationNode. CheckCreate inserts both (check.rs) and CheckCash removes both, but CheckCancel removed neither: it deleted the object and adjusted the reserve, leaving two dangling directory entries.

Independence program

No independence record for UTC day 2026-07-24; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Uniform-Price Batch Auction Offers** (dex) Add a tfBatchAuction flag to OfferCreate marking an offer as deferred from immediate sequential crossing. During ledger close, a new post-transaction
- **Net-Settlement Mode for Batch Transactions** (settlement) Add an sfNetSettle flag to the XLS-56 Batch transaction envelope. When set, doApply groups the batch's inner Payment transactions by currency/issuer,
- **On-Ledger Vault Reserve Attestation** (audit) Add a new ttVAULT_ATTEST transaction, signed by a SingleAssetVault's operator, carrying an sfLiabilitiesTotal amount and an sfLiabilitiesHash (Merkle

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
07-24 13:46	—	xrpl-validator	`609c827`	fix(xrpl-ledger): CheckCash requires exactly one of Amount / DeliverMin
07-24 14:47	—	xrpl-validator	`3a718aa`	fix(xrpl-ledger): a Payment to a pseudo-account is tecNO_PERMISSION
07-24 21:01	—	xrpl-validator	`1de6d87`	fix(xrpl-ledger): OfferCreate crossing spends only accountFunds, not TakerGets
07-24 21:45	—	xrpl-validator	`34fca11`	fix(xrpl-ledger): honour keepRoot when a directory page empties
07-24 21:45	—	xrpl-validator	`b79f40f`	fix(xrpl-ledger): CheckCancel unlinks the Check from both owner directories

Times in UTC. A commit appears on the day it was committed to main or merged into it.